

Lucas Robinet

Ph.D. Student

Toulouse, France

French citizenship

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Education

- **Université Paul Sabatier** **Toulouse, FR**
Ph.D. in Multimodal Deep Learning for brain tumor analysis *Expected 11/2025*
 - **ISAE Supaero** **Toulouse, FR**
M.Sc. in Mathematics, Computer Science and Deep Learning *09/2018 - 08/2022*
 - **Université Paul Sabatier** **Toulouse, FR**
M.Sc. in Applied Mathematics and Operations Research *09/2021 - 08/2022*
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Experiences

- **Institute of Technological Research Saint Exupery** **Toulouse, FR**
Ph.D. Student in Multimodal Deep Learning for brain tumor analysis *11/2022 - 11/2025*
 - Development of robust multimodal deep learning frameworks for brain tumor analysis, with a focus on handling missing data across MRI, pathology, and omics modalities
 - Application of these methods to address key scientific questions, including biomarker prediction, patient prognosis and tumor segmentation
 - Exploration of cross-modal integration strategies (radiology–biology and pathology–omics) using self-supervised learning to advance precision medicine in glioma*Deep Learning Intern* *04/2022 - 11/2022*
 - Investigation of the feasibility of predicting MGMT promoter status from multimodal MRI of glioblastoma patients with confidence metrics and explainability, concluding that robust prediction was not achievable**Technology:** Python, PyTorch, scikit-learn, Docker, Amazon Web Services
 - **Airbus Defence and Space** **Toulouse, FR**
Deep Learning Consultant *09/2021 - 03/2022*
 - Implementation of a MLOps automated pipeline to retrieve new satellite data, manage dataset versioning, fine-tune models, and deploy updated versions to production*Deep Learning Intern* *02/2021 - 07/2021*
 - Development and deployment of computer vision algorithms for satellite imagery, including object detection and classification**Technology:** Python, PyTorch, Google Cloud Platform, Docker, DVC, ZenML
 - **Epsilone S.A.S** **Toulouse, FR**
Data Scientist Intern *08/2020 - 01/2021*
 - Development of machine learning and deterministic algorithms to identify aerodynamic and rotor imbalances in wind turbines
 - Design and automation of reporting pipelines for potential imbalances, providing structured PDF reports to customers**Technology:** Python, Tensorflow, scikit-learn, Pandas
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Publications

- Masked Omics Modeling for Multimodal Representation Learning across Histopathology and Molecular Profiles 2025
Submitted to Medical Image Analysis
- Multimodal Masked Autoencoder Pre-training for 3D MRI-Based Brain Tumor Analysis with Missing Modalities 2025
Submitted to Artificial Intelligence in Medicine
- DRIM: Learning Disentangled Representations from Incomplete Multimodal Healthcare Data 2024
*Medical Image Computing and Computer Assisted Intervention, **Spotlight MICCAI***
- IMITATE: Image Registration with Context for Unknown Time Frame Recovery 2024
*IEEE International Symposium on Biomedical Imaging, **Oral ISBI***
- MRI-based Deep Learning Tools for MGMT Promoter Methylation detection 2023
Cancers, Journal Article
- Reviewer: AAAI 2026, MICCAI 2025, MIDL 2025

Teaching

- ISAE Supaero, Master 2, Machine learning lecturer and teaching assistant 2022 - 2025
- Toulouse Tech, Master 2, Machine learning lecturer and teaching assistant 2022 - 2023

Side work & distinctions

- **MONAI Contributor** - Python library for AI in Medical Imaging with $\approx 7k$ stars 2023 - now
- Winner of the International VisioMel Hackathon ($2^{th}/542$) - Predicting Melanoma Relapse 2023
- **Community** - Moderator of a French AI chat room on discord with over 3000 members 2023 - now

Skills

- **Programming**
Experienced in Python.
- **Machine Learning & Deep Learning Frameworks**
Extensive experience with PyTorch and scikit-learn.
- **Software Engineering**
Open Source Development, Python Packaging, Cloud Computing, Docker, healthy habits regarding code quality.
- **Languages**
Fluent French. Professional English. Intermediate Italian.

Hobbies

- Sciences
- Alpine skiing
- Tennis
- Philosophy (fresh)
- Soccer
- Programming